

Paper Title/Citation	Method used	Metrics Used	Results	Gaps
MDPI: Attallah, O. ADHD-AID: Aiding Tool for Detecting Children's Attention Deficit Hyperactivity Disorder via EEG-Based Multi-Resolution Analysis and Feature Selection. Biomimetics 2024, 9, 188. https://doi.org/10.3390/biomimetics9030188	→ Pre-processing: band-pass IIR Butterworth filter of order 6 with a frequency range from 0.5 Hz to 60 Hz. Afterward, a 50 Hz IIR notch filter of order 2. Each EEG signal is segmented into four-second intervals → Multi-Resolution Analysis and Feature Ext: VDM, DWT and EWT. Extract statistical numbers, Concatenate into one massive list. Sequential forward strategy to search for the best combination of sites. → Feature Selection(analysis of variance): ANOVA, chi squared, Kruskal Wallis → ML Classifier: C-SVM, Q-SVM, M-SVM, kNN, and ANN. (k fold validation, k = 10)	→ Performance metrics: Detection accuracy, confusion matrix, Sensitivity, Specificity, Precision, F1-Score, MCC, along with ROC, and AUC.	→ Multi-Domain Feature Extraction Results: Ran all ML models using features from only DWT or EWT or VDM, Q-SVM, and C-SVM than M-SVM, kNN, ANN perform better with time domain features and DWT than other fs methods. → Electrode Placement Site Results: Merge all features and run all ML Models against electrode placement sites. → Electrode Site Selection Results: Results of the Sequential Forward Search of multiple electrode placements. → FS Methods: Compare ML models with different FS methods and number of features using detection accuracy, confusion matrix, Sensitivity, Specificity, Precision, F1-Score, MCC, along with ROC, and AUC.	→ As stated in the paper, the author did not use DL, XAI and uncertainty quantification methods. → The author used the rigid boundaries of brainwaves, whereas EEG research calculates Individual Alpha Frequency (IAF). For example, if your IAF is 9 Hz, your custom Theta band might be shifted down to 3–7 Hz.
MDPI: Bansal, J.; Gangwar, G.; Aljaidi, M.; Alkoradees, A.; Singh, G. EEG-Based ADHD Classification Using Autoencoder Feature Extraction and ResNet with Double Augmented Attention Mechanism. Brain Sci. 2025, 15, 95. https://doi.org/10.3390/brainsci15010095	→ Pre-processing: Bandpass filter was used to keep the frequencies between 4 and 40 Hz. Plot PSD on both before and after filtering. ICA were used to remove the EEG artifacts. SNR and power spectral analysis to analyse signal quality. → Feature Extraction: Autoencoder - Encoder featured 3 fully connected layers 128, 64, and 32 neurons, respectively, and using the ReLU activation function. 32 neurons represent the latent space used for feature selection using Reptile search algorithm. The loss function used was MSE after MAE and Huber loss proved to be inadequate. → Model: ResNet with 18 residual layers and a dual augmented attention mechanism.	→ Performance Metrics: Accuracy, precision, recall, F1, AUC, specificity and sensitivity.	Out of 10,000 samples, the model demonstrates exceptional performance with 4961 records correctly identified as class 0 (True Negatives) and 4981 records accurately classified as class 1 (True Positives). Misclassifications are minimal, with only 49 False Positives (class 0 misclassified as class 1) and 9 False Negatives (class 1 misclassified as class 0).	→ As stated in the paper, there is a need for deep learning models for biological applications which this paper does not address. → Moreover, the children had been taking Ritalin 6 months prior to the data collection which introduced multiple confounds. → ADHD involves multiple brain regions and their activity, this paper has focused only on the visual aspect which doesn't capture all the other aspects affected because of ADHD such as working memory, auditory function, etc.
Springer Nature: Hassan, U., & Singhal, A. (2024). Convolutional neural network framework for EEG-based ADHD diagnosis in children. Health information science and systems, 12(1), 44. https://doi.org/10.1007/s13755-024-00305-7 Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC11365922/	→ Pre-processing: A band-pass filter of order 4 with cut-off frequency from 2Hz and 47Hz followed by segmentation. → CCA: Used to find linear relations between EEG data. → CNN Architecture: The first layer is a temporal convolutional layer with a kernel size if 64, the second layer has a separate convolution with 32 filters and a kernel size of 16. The output layer uses sigmoid function.	5 fold cross validation scheme was used along with performance metrics like accuracy, specificity and sensitivity.	→ Frontal channels exhibit higher accuracy (more than 95%). → The authors also compared the performance metrics with CCA and without it, and the system performed better with wrt accuracy, sensitivity and soecificity.	
Elsevier: Behnam Latifi, Ali Amini, Ali Motie Nasrabadi. Siamese based deep neural network for ADHD detection using EEG signal, Computers in Biology and Medicine, Volume 182,2024, 109092, ISSN 0010-4825, https://doi.org/10.1016/j.combiomed.2024.109092	Need Access			

Elsevier: Behrad TaghiBeyglou, Ashkan Shahbazi, Fatemeh Bagheri, Sina Akbarian, Mehran Jahed, Detection of ADHD cases using CNN and classical classifiers of raw EEG, Computer Methods and Programs in Biomedicine Update, Volume 2, 2022, 100080, ISSN2666-9900, https://doi.org/10.1016/j.cmpbup.2022.100080.	The authors have used a modified version of Mohammadi/Nasarbadi dataset, where the sampling rate is 512Hz. The training set contains 196 ADHD and 132 control epochs from one group of subjects, while the validation set contains 37 ADHD, 35 healthy epochs from the other group of subjects. → CNN Architecture: The network is fed pre-processed EEG signals and use 2 spatial blocks and 2 temporal blocks. The first spatial layer uses 16 filters with a size of (10 x 1) and the second layer uses 16 filters but with a size of (4 x 1) to compute weighted average between different channels. The first temporal block has 32 filters, size (1×128) to understand the time-series relationship and the second temporal block with with 32 filters and the size of (1×64). The final block has a single neuron with a sigmoid activation acts as the classification block. Except the final layer, all other layers use ReLU activation. → Classical ML: The paper uses a hybrid approach where the last dense layer features are fed into classical ML models -- NLSVM, logistic regression, Random forest, Gaussian naive bayes, k NN	→ Performance Metrics: Accuracy, precision, recall, F1, specificity and sensitivity.		
Elsevier: Mohammadreza Bakhtyari, Sayeh Mirzaei, ADHD detection using dynamic connectivity patterns of EEG data and ConvLSTM with attention framework, Biomedical Signal Processing and Control, Volume 76, 2022, 103708, ISSN 1746-8094, https://doi.org/10.1016/j.bspc.2022.103708.	Need Access			
IEEE Xplore: M. Maniruzzaman, M. A. M. Hasan, N. Asai and J. Shin, "Optimal Channels and Features Selection Based ADHD Detection From EEG Signal Using Statistical and Machine Learning Techniques," in IEEE Access, vol. 11, pp. 33570-33583, 2023, doi: 10.1109/ACCESS.2023.3264266. Link: https://ieeexplore.ieee.org/document/10091501	→ Feature Extraction: Time domain, morphological and non-linear features were extracted. → Channel Selection: Independent T - test and SVM were used. → Feature Selection: LASSO-LR → Classification Models: Different classification models were used: 1. Gaussian Process Classification with different kernel models like radial basis function (RBF), dot product, and rational quadratic kernel. 2. Random Forest: The parameters were tuned using grid search algorithm. 3. k NN: Euclidean distance was used. 4. Multi-layer Perceptron: The hyper-parameters were tuned using grid search algorithm. 5. Decision Tree: The parameters were tuned using grid search algorithm. 6. Logistic Regression	Five performance metrics such as accuracy (ACC), recall (Rec), precision (Prec), F1-score, and area under the curve (AUC) computed from the ROC curve were used to evaluate the performance of predictive models.	→ GP with RBF-based classifier gave comparatively better performance scores for the selected channels than other classifiers. → The authors also compared model performance with and without channel selection, and it was observed that a with channel classifier performed better than the one without it.	→ The authors state the need of extension using a deep learning-based classifier and try to develop a web-based method for the automated prediction.
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